

Supervised Learning

Generalized Linear Models

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Mathematical Prerequisites

- **Multinomial distribution** is a generalization of the binomial distribution
For example, it models the probability of counts of each side for rolling a k -sided die n times
- For n independent trials each of which leads to a success for exactly one of k categories
- Multinomial distribution gives the probability of any particular combination of numbers of successes for the various categories.

- For $k = 2$ and $n = 1$, the multinomial distribution is the **Bernoulli distribution**
- When $k = 2$ and $n > 1$, it is the **binomial distribution**
- When $k > 2$ and $n = 1$, it is the **categorical distribution**

Generalized Linear Models

- So far we have seen regression example and classification example
- In regression example, we had $y|x; \theta \sim \mathcal{N}(\mu, \sigma^2)$, and in the classification $y|x; \theta \sim \text{Bernoulli}(\phi)$ for some appropriate definition of μ and ϕ
- Both of these methods are special cases of a broader family of models, called Generalized Linear Models (GLMs)
- GLMs extend the linear modeling framework to variables that are not Normally distributed

Exponential Family

- A class of distribution is in the exponential family if it can be written in the form

$$p(y; \eta) = b(y) \exp(\eta^T T(y) - a(\eta))$$

where

- η is called **natural parameter** of the distribution
- $T(y)$ is the **sufficient statistics** (in most of our cases, $T(y) = y$)
- $a(\eta)$ is the **log partition function**
- Quantity $e^{-a(\eta)}$ plays the role of a normalization constant, that ensures the distribution $p(y; \eta)$ sums/integrates over y to 1
- Fixed choice of T, a and b defines a family of distribution that is parameterized by η , as we vary η , we get different distributions within this family

Bernoulli and Gaussian distributions are examples of exponential distribution

- Bernoulli distribution with mean ϕ , $Bernoulli(\phi)$, specifies a distribution over $y \in \{0, 1\}$ so that $p(y = 1; \phi) = \phi$ and $p(y = 0; \phi) = 1 - \phi$
- As we vary ϕ , we obtain Bernoulli distribution with different means
- We write the Bernoulli distribution as

$$\begin{aligned} p(y; \phi) &= \phi^y (1 - \phi)^{1-y} \\ &= \exp(\log(\phi^y (1 - \phi)^{1-y})) \\ &= \exp(\log \phi^y + \log(1 - \phi)^{(1-y)}) \\ &= \exp(y \log \phi + (1 - y) \log(1 - \phi)) \\ &= \exp(y \log \phi + \log(1 - \phi) - y \log(1 - \phi)) \\ &= \exp(y \log \phi - y \log(1 - \phi) + \log(1 - \phi)) \\ &= \exp\left(y \log \frac{\phi}{1 - \phi} + \log(1 - \phi)\right) \end{aligned}$$

- Comparing $p(y; \phi) = \exp\left(y \log \frac{\phi}{1 - \phi} + \log(1 - \phi)\right)$ with

$$p(y; \eta) = b(y) \exp(\eta^T T(y) - a(\eta))$$

we get following parameters of exponential distribution

- $\eta = \log\left(\frac{\phi}{1 - \phi}\right)$ (we can also write $\phi = \frac{1}{1 + e^{-\eta}}$)
- $T(y) = y$
- $a(\eta) = -\log(1 - \phi) = \log(1 + e^\eta)$
- $b(y) = 1$
- This shows that Bernoulli distribution is a special case of exponential distribution

Consider Gaussian distribution

- While deriving linear regression, the value of σ^2 had no effect on our final choice of θ and $h_\theta(x)$
- To simplify derivation, assume $\sigma^2 = 1$

$$p(y; \mu) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}(y - \mu)^2\right)$$

- $$\begin{aligned} &= \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}(y^2 - 2\mu y + \mu^2)\right) \\ &= \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}y^2\right) \cdot \exp\left(\mu y - \frac{1}{2}\mu^2\right) \end{aligned}$$
- On comparing with $p(y; \eta) = b(y)\exp(\eta^T T(y) - a(\eta))$, we get
 - $b(y) = (1/\sqrt{2\pi}) \exp(-y^2/2)$
 - $\eta = \mu$
 - $T(y) = y$
 - $a(\eta) = \mu^2/2 = \eta^2/2$

Constructing GLMs

Problem Scenario

Suppose you would like to build a model to estimate the number y of customer arriving in your store (or number of page-views on your websites) in any given hour, based on certain features x such as store promotions, recent advertising, weather, day-of the week, etc. We know that the Poisson distribution gives good model for number of visitors. Knowing this, how can we come up with a model for our problem?

- Is there a general recipe for constructing models in which y (given x and θ) comes from any exponential distribution?
- Yes, it is possible for any exponential distribution

- Consider a classification or regression problem where we would like to predict the value of y as a function of x
- To derive a GLM for this problem, we make following assumptions about the condition distribution of y given x and about our model:
 - ① $y|x; \theta \sim \text{ExponentialFamily}(\eta)$ i.e given x and θ , the distribution of y follows some exponential family distribution, with parameter η
 - ② Given x , our goal is to predict the expected value of $T(y)$ given x . In most of our cases $T(y) = y$, this means we would like the prediction $h(x)$ output by learned hypothesis h to satisfy $h(x) = E[y|x]$
 - ③ The natural parameter η and the inputs x are related linearly, $\eta = \theta^T x$ (or, $\eta_i = \theta_i^T x$, if η is vector valued)

- These assumption/design choices allow us to derive a very elegant class of learning algorithms, namely GLMs, that have many desirable properties
- Resulting models are very effective for modelling different types of distribution over y
- Both least squares and logistic regression can be derived as GLMs

Ordinary Least Squares

- Consider the setting where the target variable y is continuous, and $y|x \sim \mathcal{N}(\mu, \sigma^2)$
- In the formulation of the Gaussian as an exponential family distribution, we have $\eta = \mu$, so we have

$$\begin{aligned}h(x) &= E[y|x; \theta] \quad (\text{from A2}) \\&= \mu \quad (\text{follows from fact that } y|x; \theta \sim \mathcal{N}(\mu, \sigma^2)) \\&= \eta \quad (\text{from A1}) \\&= \theta^T x \quad (\text{from A3})\end{aligned}$$

Logistic Regression

- Given that $y \in \{0, 1\}$, $y|x; \theta \sim \text{Bernoulli}(\phi)$
- In our formulation of Bernoulli distribution as exponential family, we had $\phi = \frac{1}{(1+e^{-\eta})}$
- If $y|x; \theta \sim \text{Bernoulli}(\phi)$, then $E[y|x; \theta] = \phi$, so we get

$$\begin{aligned}h(x) &= E[y|x; \theta] \\&= \phi \\&= \frac{1}{(1 + e^{-\eta})} \\&= \frac{1}{(1 + e^{-\theta^T x})}\end{aligned}$$

- Function g giving the distribution's mean as a function of the natural parameter ($g(\eta) = E[T(y; \eta)]$) is called **canonical response function**
- Inverse of canonical response function, g^{-1} , is called the **canonical link function**

Softmax Regression

- Softmax regression (or multinomial logistic regression) is a generalization of logistic regression to the case where we want to handle multiple classes
- In logistic regression we assumed that labels were binary: $y^{(i)} \in \{0, 1\}$
- Softmax regression allows us to handle $y^{(i)} \in \{1, \dots, k\}$, where k is the number of classes
- We'll model it as distributed according to a multinomial distribution
- Lets derive GLM for modelling this type of multinomial data
- We begin by expressing multinomial as an exponential family distribution

- To parameterize a multinomial over k possible outcomes, one could use k parameters $\phi_1, \dots, \phi_k$ specifying probability for each outcome
- We'll parameterize the multinomial with only $k - 1$ parameters, $\phi_1, \dots, \phi_{k-1}$ where

$$\phi_i = p(y = i; \phi), \text{ and } p(y = k; \phi) = 1 - \sum_{i=1}^{k-1} \phi_i$$

- To express the multinomial as an exponential family distribution, we'll define $T(y) \in \mathbb{R}^{k-1}$ as follows

$$T(1) = \begin{bmatrix} 1 \\ 0 \\ \dots \\ 0 \end{bmatrix}, T(2) = \begin{bmatrix} 0 \\ 1 \\ \dots \\ 0 \end{bmatrix}, \dots, T(k-1) = \begin{bmatrix} 0 \\ 0 \\ \dots \\ 1 \end{bmatrix}, T(k) = \begin{bmatrix} 0 \\ 0 \\ \dots \\ 0 \end{bmatrix}$$

- We write $(T(y))_i$ to denote the i th element of vector $T(y)$
- An indicator function $1\{\cdot\}$ takes on a value of 1 if its argument is true, and 0 otherwise
- For example, $1\{2 = 3\} = 0$, and $1\{2 = 4 - 2\} = 1$
- Now, we can write the relationship between $T(y)$ and y as $(T(y))_i = 1\{y = i\}$
- We also have, $E[(T(y))_i] = p(y = i) = \phi_i$
- We'll show that the multinomial is a member of the exponential family

$$\begin{aligned}
p(y; \phi) &= \phi_1^{1\{y=1\}} \phi_2^{1\{y=2\}} \dots \phi_k^{1\{y=k\}} \\
&= \phi_1^{1\{y=1\}} \phi_2^{1\{y=2\}} \dots \phi_k^{1 - \sum_{i=1}^{k-1} 1\{y=i\}} \\
&= \phi_1^{(T(y))_1} \phi_2^{(T(y))_2} \dots \phi_k^{1 - \sum_{i=1}^{k-1} (T(y))_i} \\
&= \exp \left(\log \left(\phi_1^{(T(y))_1} \phi_2^{(T(y))_2} \dots \phi_k^{1 - \sum_{i=1}^{k-1} (T(y))_i} \right) \right) \\
&= \exp \left((T(y))_1 \log(\phi_1) + (T(y))_2 \log(\phi_2) + \dots + \left(1 - \sum_{i=1}^{k-1} (T(y))_i \right) \log(\phi_k) \right) \\
&= \exp((T(y))_1 \log(\phi_1/\phi_k) + (T(y))_2 \log(\phi_2/\phi_k) + \dots + (T(y))_{k-1} \log(\phi_{k-1}/\phi_k) + \log \phi_k) \\
&= b(y) \exp \left(\eta^T T(y) - a(\eta) \right)
\end{aligned}$$

where $\eta = \begin{bmatrix} \log(\phi_1/\phi_k) \\ \log(\phi_2/\phi_k) \\ \vdots \\ \log(\phi_{k-1}/\phi_k) \end{bmatrix}$

$a(\eta) = -\log(\phi_k)$

$b(y) = 1$

- The link function is given by

$$\eta_i = \log \frac{\phi_i}{\phi_k}$$

- Invert the link function and derive the response function

$$\begin{aligned}
e^{\eta_i} &= \frac{\phi_i}{\phi_k} \\
\phi_k e^{\eta_i} &= \phi_i \\
\phi_k \sum_{i=1}^k e^{\eta_i} &= \sum_{i=1}^k \phi_i = 1 \\
\phi_k &= 1 / \sum_{i=1}^k e^{\eta_i}
\end{aligned}$$

Hence the response function is given by

$$\phi_i = \frac{e^{\eta_i}}{\sum_{j=1}^k e^{\eta_j}}$$

Function mapping from the η 's to the ϕ 's is called the **softmax** function

- As per A3 of GLM construction,
 $\eta_i = \theta_i^T x, \quad \forall i = 1, \dots, k-1$, where $\theta_1, \dots, \theta_{k-1} \in \mathbb{R}^{n+1}$ are the parameters of our model
- For notational convenience, $\phi_k = 0$ so that $\eta_k = \theta_k^T x = 0$
- The conditional distribution of y given x is given by

$$\begin{aligned} p(y = i|x; \theta) &= \phi_i \\ &= \frac{e^{\eta_i}}{\sum_{j=1}^k e^{\eta_j}} \\ &= \frac{e^{\theta_i^T x}}{\sum_{j=1}^k e^{\theta_j^T x}} \end{aligned}$$

- This model, which applies to classification problem where $y \in \{1, \dots, k\}$ is called **softmax regression**
- Softmax regression is the generalization of logistic regression

Our hypothesis function will output

$$\begin{aligned} h(x) &= E[T(y)|x; \theta] \\ &= E \left[\begin{array}{c|c} \begin{matrix} 1\{y=1\} \\ 1\{y=2\} \\ \vdots \\ 1\{y=k-1\} \end{matrix} & \begin{matrix} | \\ | \\ |x; \theta \\ | \end{matrix} \end{array} \right] \\ &= \begin{bmatrix} \phi_1 \\ \phi_2 \\ \vdots \\ \phi_{k-1} \end{bmatrix} \\ &= \begin{bmatrix} \frac{e^{\theta_1^T x}}{\sum_{j=1}^k e^{\theta_j^T x}} \\ \frac{e^{\theta_2^T x}}{\sum_{j=1}^k e^{\theta_j^T x}} \\ \vdots \\ \frac{e^{\theta_{k-1}^T x}}{\sum_{j=1}^k e^{\theta_j^T x}} \end{bmatrix} \end{aligned}$$

- Hypothesis $h(x)$, will output the estimated probability that $p(y = i|x; \theta)$ for every $i = 1, \dots, k$
- $p(y = k|x; \theta)$ can be calculated as $1 - \sum_{i=1}^{k-1} \phi_i$
- For a set of training example, $\{(x^{(i)}, y^{(i)}) | i = 1, \dots, m\}$ we would like to learn the parameters θ_i for this model
- The log-likelihood function is computed as

$$\begin{aligned}
 L(\theta) &= \prod_{i=1}^m p(y^{(i)} | x^{(i)}; \theta) \\
 l(\theta) &= \log \left(\prod_{i=1}^m p(y^{(i)} | x^{(i)}; \theta) \right) \\
 &= \sum_{i=1}^m \log p(y^{(i)} | x^{(i)}; \theta) \\
 &= \sum_{i=1}^m \log \prod_{l=1}^k \left(\frac{e^{\theta_l^T x^{(i)}}}{\sum_{j=1}^k e^{\theta_j^T x^{(i)}}} \right)^{1_{\{y^{(i)}=l\}}}
 \end{aligned}$$

- MLE of the parameter can be obtained by maximizing $l(\theta)$ in terms of θ using method such as gradient ascent or Newton's method